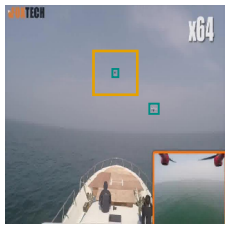
Direct-parent module contrasts | matched 200-epoch seed-1 checkpoints

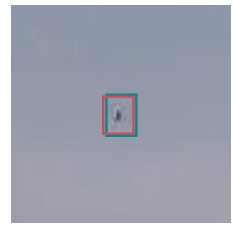
GT: teal solid box Prediction: red solid box Crop: amber Display confidence ≥0.25

a P2: small-target score rescue (20 px)



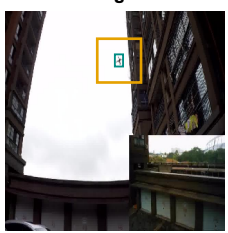


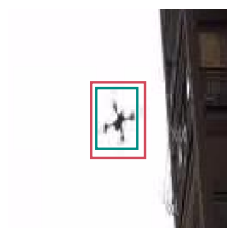
YOLO11n: No IoU≥0.50 detection



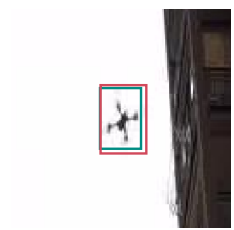
+P2: IoU 0.75 | conf 0.39

b ADown: tighter small-target box (29 px)



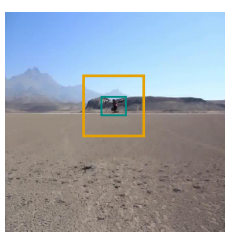


+P2: IoU 0.61 | conf 0.57



+P2+ADown: IoU 0.78 | conf 0.57

c ADown: confidence-collapse counterexample (62 px)





+P2: IoU 0.78 | conf 0.55



+P2+ADown: No IoU≥0.50 detection

Same native-pixel crop within each row; no image enhancement. Post-hoc examples, not representative samples.